Anushka Dongaonkar

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Summary

MS Bioinformatics graduate with experience in next-generation sequencing (NGS) data analysis, computational pipeline development, and multi-omics data integration. Skilled in Python, R, SQL, Linux, and HPC environments for analyzing bulk and single-cell sequencing data. Interested in developing computational methods and collaborating with experimental biologists to generate biological insights.

Education

MS in Bioinformatics | **Boston University, Boston** — Sep 2024 – Jan 2026

GPA – 3.93/4

Bachelors in Biotechnology | **Vellore Institute of Technology, India** — Sep 2020 - May 2024

GPA - 8.75/10

Technical Skills

- **Programming:** Python, R, SQL, MATLAB, Linux, Bash, HPC, Nextflow
- **Machine Learning & Statistics:** Regression, classification, clustering, dimensionality reduction, gradient boosting, cross-validation, supervised and unsupervised machine learning
- **Biological Data Types:** Bulk RNA-seq, scRNA-seq, transcriptomics, epigenomics, proteomics, metabolomics, multi-omics datasets
- **NGS & Bioinformatics Tools:** STAR, BWA, Bowtie2, ScanPy, Seurat, DESeq2, edgeR, MACS2, HOMER, Bioconductor, Biopython, scikit-learn
- **Genomic Formats & Resources:** FASTQ, BAM, VCF, BED, Ensembl, UCSC Genome Browser, NCBI, GenBank
- **Web & Application Development:** Flask, Docker, R Shiny, HTML, CSS, JavaScript

Relevant Experience

***Web Developer* | Wunderlich Lab, Boston University** — Sep 2025 – Dec 2025

Designed computational infrastructure and software tools supporting analysis and exploration of regulatory genomics datasets generated from STARR-seq experiments in *Drosophila melanogaster*.

- Developed a relational SQL database containing ~35,000 enhancer–gene associations derived from STARR-seq experiments in *Drosophila melanogaster.*
- Built a Flask-based web application enabling researchers to dynamically query enhancer–gene relationships and visualize regulatory genomic data.
- Implemented interactive front-end components using HTML, CSS, and JavaScript to support flexible data exploration.
- Improved accessibility and usability of large regulatory genomics datasets within the lab by reducing manual dataset navigation and enabling structured queries.

***Research Assistant* | Zhang Lab, Boston University** — Oct 2024 – Sep 2025

Conducted computational analysis of large-scale proteomics and metabolomics datasets to identify biomarkers associated with cognitive decline using reproducible Python and R workflows.

- Integrated large-scale proteomics and metabolomics datasets to identify candidate biomarkers associated with Mild Cognitive Impairment.
- Built reproducible analytical workflows in Python and R for data preprocessing, quality control, normalization, and downstream statistical analysis.
- Applied supervised machine learning models including Random Forest, XGBoost, Logistic Regression and OPLS-DA to classify disease states and evaluate predictive performance.
- Identified 15 candidate metabolite and protein biomarkers associated with disease progression and evaluated their predictive utility using supervised machine learning models, achieving AUROC ~0.85.
- Performed feature selection and model evaluation using cross-validation and comparative model benchmarking.

***Teaching Assistant* | Questrom School of Business, Boston University** — Jan 2025 – Mar 2025

Supported instruction for graduate coursework in unsupervised and unstructured machine learning.

- Assisted in developing and grading assignments related to clustering, dimensionality reduction, and natural language processing.
- Assisted students with implementation of machine learning methods and interpretation of model outputs.

***Research Trainee* | ABOpen Lab, CSIR-Institute of Microbial Technology, India** — Jan 2024 – June 2024

Worked on computational drug repurposing strategies targeting bacterial pathogens.

- Implemented and scaled a network-based drug repurposing workflow for *Pseudomonas aeruginosa*, screening 792,612 protein–drug interactions to identify potential therapeutic candidates.
- Identified 383 candidate drugs through network-based prioritization.
- Developed Python scripts to perform data analysis and workflow automation on an HPC cluster environment.

***Clinical Research Intern* | NeuroGen Brain and Spinal Research Institute, India** — May 2023 – June 2023

- Analyzed clinical datasets from 36 patient samples to evaluate the effect of integrative therapies on biological aging markers.
- Performed statistical analysis using t-tests to assess treatment efficacy.
- Improved documentation and data organization between research and clinical teams.

***Research Intern* | Protein Biophysics Lab, IIT-Madras, India** — May 2023 – Sep 2023

- Investigated the relationship between protein size and radius of gyration using protein structural data from the PDB database.
- Implemented statistical analysis and visualization workflows in MATLAB to characterize structural trends in protein datasets.

Bioinformatics Projects

Single-cell RNA-seq Analysis with ScanPy [GitHub] Apr 2025 – May 2025

Implemented an end-to-end single-cell transcriptomics analysis workflow using ScanPy.

- Processed over 83,000 cells from primary tumors, liver metastases, and adjacent normal tissue using ScanPy.
- Developed an end-to-end workflow including quality control, normalization, batch integration, dimensionality reduction, Leiden clustering, differential expression analysis, and cell-type annotation.
- Performed trajectory inference to investigate transcriptional state transitions and characterize tumor-associated immune populations.
- Annotated cell populations using canonical marker genes and validated annotations against published literature.

Gut Microbiome Disease Prediction using Machine Learning [GitHub] Sep 2025 – Dec 2025

Analyzed gut microbiome profiles from 1,831 subjects to investigate associations between microbial composition and cardiometabolic disease status using R and Python.

- Processed and analyzed microbiome abundance data using taxonomic filtering, relative abundance normalization, CLR transformation, and diversity analysis.
- Computed alpha diversity metrics (Shannon, Simpson, richness) and performed beta diversity analysis and PERMANOVA to characterize microbiome differences between healthy and disease cohorts.
- Integrated microbiome features with clinical metadata and developed machine learning models (Random Forest, Logistic Regression, XGBoost) for disease classification, achieving ROC-AUC up to 0.89.

Bulk RNA-seq Data Analysis Pipeline [GitHub] Feb 2025 – Mar 2025

Built a reproducible RNA-seq analysis workflow for gene expression analysis.

- Implemented a pipeline including quality control, read alignment, expression quantification, differential expression, and pathway enrichment analysis.
- Analyzed gene expression patterns across multiple pancreatic differentiation stages comparing TYK2 knockout and wild-type conditions.
- Replicated results from the original study and identified KRAS upregulation and disrupted endocrine function associated with TYK2 perturbation.

ChIP-seq Data Analysis and Motif Discovery [GitHub] Mar 2025 – Apr 2025

Developed an end-to-end epigenomic analysis pipeline for transcription factor binding analysis.

- Implemented a workflow including alignment, peak calling, genomic annotation, and motif enrichment analysis.
- Identified RUNX1 binding sites and analyzed their distribution across promoters and regulatory regions.
- Explored potential regulatory roles of RUNX1 in breast cancer transcriptional programs.

Data Analysis and Visualization Application [Link] Nov 2024 – Dec 2024

- Developed an R Shiny application enabling interactive exploration of gene expression datasets.
- Built visualizations allowing users to analyze 60,000 genes across 32 Alzheimer's disease patient samples.
- Designed dynamic plots and interactive filters to support exploratory analysis of transcriptomic datasets.

Publications

Evaluating the Impact of Anti-Aging Integrative Therapy on Cellular Aging and Quality of Life [link] Oct 2024

Alok Sharma, Hemangi M. Sane ... , ***Anushka Dongaonkar***

Sono-assisted adsorption of nitrate by Fe-modified chitosan–zeolite composite material [link] Mar 2024

Gopal Italiya, ***Anushka Dongaonkar****, Nithyashree Jayakumar, Sangeetha Subramanian*